PRODUCT USER MANUAL

For Ocean Nutrient and Carbon profiles vertical distribution Product

MULTIOBS_GLO_BGC_NUTRIENTS_CARBON_PROFILES_MYNRT_015_009 (CANYON)

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
0.0	January 2019	all	Initial version of the document	R. Sauzède, H. Claustre	
1.0	December 2019		Minor updates linked to the temporal extension of the time series	R. Sauzède, H. Claustre	C. Derval
2.0	January 2021		Minor updates linked to the temporal extension of the time series and the addition of Mediterranean Sea profiles	R. Sauzède, H. Claustre	
3.0	November 2022		Minor updates: the temporal extension of Nutrient profiles up to August 2022 Major updates: the release of new carbonate system variables (total alkalinity, dissolved inorganic carbon, pH and partial pressure of carbon dioxide) New name of the product and new template	R. Sauzède, P.R. Renosh, H.Claustre	Copernicus Marine Service Manageme nt
4.0	November 2023		Set up of the Operational Temporal Extension	R. Sauzède, P.R. Renosh, H.Claustre	H.Etienne
5.0	November 2024		Major updates: transition to near real-time product (up to Month-1) with monthly data with associated refined uncertainties.	P.R. Renosh, R. Sauzède, H. Claustre	H.Etienne

Table of Contents

GLOSSARY AND ABBREVIATIONS *4*

DATA ACCESS **5**

I INTRODUCTION *6*

I.1 Summary **6**

I.2 History of changes **7**

II DESCRIPTION OF THE PRODUCT SPECIFICATION *9*

II.1 General Information **9**

II.2 Details of datasets **11**

II.3 Production System Description **13**

II.4 Grid **15**

II.5 Vertical Levels **15**

II.6 Processing information **16**

II.6.1 Update Time 16

II.6.2 Temporal extent stored on delivery mechanism 16

II.6.3 Time averaging 16

III FILE FORMAT *17*

IV FILES NOMENCLATURE *18*

IV.1 Nomenclature of original files **18**

IV.2 Index file **18**

IV.3 Other information: mean centre of Products, land mask value, missing value **19**

IV.3.1 File size 19

IV.4 Structure of files **19**

V REFERENCES *20*

VI ANNEX *21*

GLOSSARY AND ABBREVIATIONS

CF	Climate Forecast (convention for NetCDF)
CMEMS	Copernicus Marine Environment Monitoring Service
GLO	Global
MULTIOBS	Multi Observations
REP	Reprocessed
NetCDF	Network Common Data Form
PUM	Product User Manual
QUID	Quality Information Document
BGC	Biogeochemical
NO_3^-	Nitrate
PO_4^{3-}	Phosphate
$Si(OH)_4$	Silicate
O_2	Dissoved oxygen
CANYON	CArbonate system and Nutrients concentration from hYdrological properties and Oxygen using a Neural-network
CONTENT	CONsisTency EstimatioN and amounT
MYNRT	Multiyear near real time
TALK	Total Alkalinity
TCO2	Dissolved inorganic carbon
pH	pH of sea water at total scale
pCO_2	Partial pressure of carbon dioxide
DGF	Direct Get File (FTP-like CMEMS service tool to download a NetCDF file)

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1 Summary

This document is the user manual for the Copernicus Marine Service nutrients (NO_3^-, PO_4^{3-} and $Si(OH)_4$) and carbonate system variables (TALK, TCO2, pH and pCO2) profiles product MULTIOBS_GLO_BGC_NUTRIENTS_CARBON_PROFILES_MYNRT_015_009. The product provides an estimation of nutrient and carbonate system variables profiles from the BGC-Argo dissolved oxygen (O_2) dataset. The profiles are aggregated by month from September 2002 to one month prior (M-1) to the present month (M). These estimations depend on the availability of concomitant profiles of pressure, temperature, salinity, with adjusted in real-time or delayed mode qualified profiles of O_2 in Argo Global Data Assembly Center (GDAC).

This product is global (see II.2) with the geographical and temporal resolutions defined from the BGC-Argo O_2 dataset.

The product is organized in 1 dataset: cmems_obs-mob_glo_bgc-nut-car_mynrt_irr_i.

It is updated every month (M) up to one month prior to the present month (M-1).

The product is published on the Copernicus Marine Service dissemination server after automatic and human quality controls. Files downloaded are in NetCDF-4 format and follow CF-1.6 convention.

The processing system is described in the Quality Information Document (**Erreur ! Source du renvoi introuvable. Erreur ! Source du renvoi introuvable.**)

I.2 History of changes

Date	Description of changes and impacted product
January 2019	1st publication of MULTIOS_GLO_BIO_REP_015_006 product.
December 2019	2nd publication of MULTIOBS_GLO_BIO_NUTRIENTS_PROFILES_REP_015_009 product. MULTIOBS_GLO_BIO_NUTRIENTS_PROFILES_REP_015_009 is the new name of MULTIOBS_GLO_BIO_REP_015_006 The time series has been completed according to new BGC-Argo-O_2 delayed mode data availability and extended until June 2019. 14 612 profiles have been added compared to previous version of the product.
January 2021	3rd publication of MULTIOBS_GLO_BIO_NUTRIENTS_PROFILES_REP_015_009 product. The time series has been completed according to new BGC-Argo-O_2 delayed mode data availability and extended until December 2020. Compared to previous version of the product, available Mediterranean Sea profiles have been added using a specifically developed regional method. A total of 53 799 profiles have been added compared to previous version of the product.
November 2022	4th publication of MULTIOBS_GLO_BGC_NUTRIENTS_CARBON_PROFILES_MYNRT_015_009 product. The time series of Nutrients has been completed according to new BGC-Argo-O_2 delayed mode data availability and extended until August 2022. There are four new carbonate system variables (A_T, DIC, pH, and pCO_2) introduced in this release. The time series extension of these carbonate system variables is coherent with Nutrient profiles. A total of 25 228 more nutrients profiles have been added compared to previous version of the product.
November 2023	5th publication of MULTIOBS_GLO_BGC_NUTRIENTS_CARBON_PROFILES_MYNRT_015_009 product.

	Temporal extension of Nutrient and Carbonate system variable profiles up to August 2023 depends upon the availability of delayed mode profiles of pressure, temperature, salinity and dissolved oxygen in Argo GDAC. Set up of Operational Temporal Extension.
November 2024	6th publication of MULTIOBS_GLO_BGC_NUTRIENTS_CARBON_PROFILES_MYNRT_015_009 product. Beginning with the November 2024 release, this product will transition to a near real-time multi-year format with monthly updates (up to approximately M-1). Unlike previous releases, the CANYON/CONTENT-based profiles will be derived from pressure, temperature, and salinity inputs from real-time, real-time adjusted, or delayed mode qualified data. In contrast, O2 input will rely exclusively on real-time adjusted or delayed mode qualified data. **FURTHERMORE, AN INDEX FILE IS PROVIDED CONTAINING ESSENTIAL INFORMATION FOR ALL PROFILES, INCLUDING FILE NAME, LONGITUDE, LATITUDE, WMO, CYCLE NUMBER, DATE, DATA MODES FOR TEMPERATURE, SALINITY AND O2, AS WELL AS THE DATE OF UPDATE STATUS. THIS FEATURE WILL ENABLE USERS TO SELECT THE MOST SUITABLE PROFILES FOR THEIR NEEDS. THE INDEX FILE WILL BE UPDATED ON A MONTHLY BASIS WITH NEW ADDITIONS OF PROFILES. THE CHANGES IN THE DATA MODE OF ANY OF THE INPUT WILL BE PROMPTLY REFLECTED IN THE INDEX FILE.**

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Information

Table 1: MULTIOBS_GLO_BGC_NUTRIENTS_CARBON_PROFILES_MYNRT_015_009 product

Product Lines	MULTIOBS_GLO_BGC_NUTRIENTS_CARBON_PROFILES_MYNRT_015_009
Geographical coverage	Global
Variables	Sea pressure, **PRES_ADJUSTED** (dbar) Quality flag of sea water pressure, **PRES_ADJUSTED_QC** Error on the sea water pressure, **PRES_ADJUSTED_ERROR** (dbar) Sea water temperature, **TEMP_ADJUSTED** (°C) Quality flag of sea water temperature, **TEMP_ADJUSTED_QC** Error on the sea water temperature, **TEMP_ADJUSTED_ERROR** (°C) Sea water salinity, **PSAL_ADJUSTED** (psu) Quality flag of sea water salinity, **PSAL_ADJUSTED_QC** Error on the sea water salinity, **PSAL_ADJUSTED_ERROR** (psu) Sea water dissolved oxygen, **DOXY_ADJUSTED** (µmol kg^{-1}) Quality flag of sea water dissolved oxygen, **DOXY_ADJUSTED_QC** Error on the sea water dissolved oxygen, **DOXY_ADJUSTED_ERROR** (µmol kg^{-1}) Nitrate concentration estimated from CANYON, **NITRATE_CANYON** (µmol kg^{-1}) Quality flag of nitrate concentration, **NITRATE_CANYON_QC** Error on the estimation of nitrate concentration, **NITRATE_CANYON_ERROR** (µmol kg^{-1}) Phosphate concentration estimated from CANYON, **PHOSPHATE_CANYON** (µmol kg^{-1}) Quality flag of phosphate concentration, **PHOSPHATE_CANYON_QC** Error on the estimation of phosphate concentration, **PHOSPHATE_CANYON_ERROR** (µmol kg^{-1}) Silicate concentration estimated from CANYON, **SILICATE_CANYON** (µmol kg^{-1}) Quality flag of silicate concentration, **SILICATE_CANYON_QC** Error on the estimation of silicate concentration, **SILICATE_CANYON_ERROR** (µmol kg^{-1}) Total alkalinity concentration estimated from CONTENT, **TALK_CONTENT** (µmol kg^{-1}) Quality flag of total alkalinity concentration, **TALK_CONTENT_QC** Error on the estimation of total alkalinity concentration, **TALK_CONTENT_ERROR** (µmol kg^{-1})

Variables	Dissolved inorganic carbon concentration estimated from CONTENT, **TCO2_CONTENT** (µmol kg^{-1}) Quality flag of dissolved inorganic carbon concentration, **TCO2_CONTENT_QC** Error on the estimation of dissolved inorganic carbon, **TCO2_CONTENT_ERROR** (µmol kg^{-1}) pH on total scale estimated from CONTENT, **PH_CONTENT (**unitless**)** Quality flag of pH, **PH_CONTENT_QC** Error on the estimation of pH, **PH_CONTENT_ERROR** (unitless) **Partial pressure of carbon dioxide estimated from CONTENT, PCO2_CONTENT** (**µatm**) Quality flag of partial pressure of carbon dioxide, **PCO2_CONTENT_QC** Error on estimation of partial pressure of carbon dioxide, **PCO2_CONTENT_ERROR** (µatm) Date of reference for Julian days, **REFERENCE_DATE_TIME** Longitude of the station, **LONGITUDE** (degree East) Latitude of the station, **LATITUDE** (degree North) Quality on position, **POSITION_QC** Julian day of the station, **JULD** Quality on date and time, **JULD_QC** Float unique identifier, **PLATFORM_NUMBER** Float cycle number, **CYCLE_NUMBER** Name of the principal investigator, **PI_NAME** List of available parameters of the station, **STATION_PARAMETERS** Coded instrument type, **WMO_INST_TYPE** Data centre in charge of float data processing, **DATA_CENTRE**
Product Type	Multi Year Near Real Time
Available time series	From September 2002 to one month (M-1) prior to the present month (M) (see online catalogue)
Temporal resolution	Profiles are aggregated in monthly NetCDF files
Target delivery time	last Thursday of the month at 14h UTC
Delivery mechanism	After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download. Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

Horizontal resolution	Discrete, corresponding to BGC-Argo O_2 profiles horizontal resolution
Number of vertical levels	from 70 to 5044 (depending on the vertical levels of the Argo profiles)
Format	NetCDF-4 CF1.6

II.2 Details of datasets

Table 2: list of the datasets and variable names and unit for the product MULTIOBS_GLO_BGC_NUTRIENTS_CARBON_PROFILES_MYNRT_015_009

MULTIOBS_GLO_BGC_NUTRIENTS_CARBON_PROFILES_MYNRT_015_009
Dataset : cmems_obs-mob_glo_bgc-nut-car_mynrt_irr_i **Variables name in the NetCDF file and Unit: Long_name & Standard_name**
PRES_ADJUSTED [decibar] **Sea Pressure** **Sea_water_pressure**
PRES_ADJUSTED_QC **Quality flag**
PRES_ADJUSTED_ERROR [decibar] **Sea water pressure, equals 0 at sea-level**
TEMP_ADJUSTED [degree_Celcius] **Sea temperature in-situ ITS-90 scale** **Sea_water_temperature**
TEMP_ADJUSTED_QC **Quality flag**
TEMP_ADJUSTED_ERROR [degree_Celcius] **Sea temperature in-situ ITS-90 scale**
PSAL_ADJUSTED [practical salinity unit] **Practical Salinity** **Sea_water_salinity**
PSAL_ADJUSTED_QC **Quality flag**
PSAL_ADJUSTED_ERROR [practical salinity unit] **Practical Salinity**
DOXY_ADJUSTED [micromole kg^{-1}] **Dissolved oxygen** **moles_of_oxygen_per_unit_mass_in_sea_water**
DOXY_ADJUSTED_QC **Quality flag**
DOXY_ADJUSTED_ERROR [micromole kg^{-1}] **Dissolved oxygen**
NITRATE_CANYON [micromole kg^{-1}] **Nitrate** **moles_of_nitrate_per_unit_mass_in_sea_water**

MULTIOBS_GLO_BGC_NUTRIENTS_CARBON_PROFILES_MYNRT_015_009
NITRATE_CANYON_QC **Quality flag**
NITRATE_CANYON_ERROR [micromole kg^{-1}] **Nitrate error**
PHOSPHATE_CANYON [micromole kg^{-1}] **Phosphate** **moles_of_phosphate_per_unit_mass_in_sea_water**
PHOSPHATE_CANYON_QC Quality flag
PHOSPHATE_CANYON_ERROR [micromole kg^{-1}] **Phosphate error**
SILICATE_CANYON [micromole kg^{-1}] Silicate **moles_of_silicate_per_unit_mass_in_sea_water**
SILICATE_CANYON_QC Quality flag
SILICATE_CANYON_ERROR [micromole kg^{-1}] **Silicate error**
TALK_CONTENT [micromole kg^{-1}] **total_alkalinity_in_sea_water**
TALK_CONTENT_QC Quality flag
TALK_CONTENT_ERROR [micromole kg^{-1}] **Total alkalinity error**
TCO2_CONTENT [micromole kg^{-1}] **dissolved_inorganic_carbon_in_sea_water**
TCO2_CONTENT_QC **Quality flag**
TCO2_CONTENT_ERROR [micromole kg^{-1}] **Dissolved inorganic carbon error**
PH_CONTENT **sea_water_ph_reported_on_total_scale**
PH_CONTENT_QC **Quality flag**
PH_CONTENT_ERROR pH **error**
PCO2_CONTENT **p**artial_pressure_of_carbon_dioxxide_in_sea_water
PCO2_CONTENT_QC **Quality flag**
PCO2_CONTENT_ERROR pCO2 error
REFERENCE_DATE_TIME **Date of reference for Julian days**
LONGITUDE [degree east] **Longitude of the station, best estimate**
LATITUDE [degree north] **Latitude of the station, best estimate**

MULTIOBS_GLO_BGC_NUTRIENTS_CARBON_PROFILES_MYNRT_015_009
POSITION_QC **Quality on position (latitude and longitude)**
JULD [days since 1950-01-01 00:00:00 UTC] **Julian day (UTC) of the station relative to REFERENCE_DATE_TIME**
JULD_QC **Quality on Date and Time**
PLATFORM_NUMBER **Float unique identifier**
CYCLE_NUMBER **Float cycle number**
PI_NAME **Name of the principal investigator**
STATION_PARAMETERS **List of available parameters for the station**
WMO_INST_TYPE **Coded instrument type**
DATA_CENTRE **Data centre in charge of float data processing**

II.3 Production System Description

The NO_3^-, PO_4^{3-}, Si(OH)4, TALK, TCO2, pH and pCO2 (NITRATE_CANYON, PHOSPHATE_CANYON, SILICATE_CANYON, TALK_CONTENT, TCO2_CONTENT, PH_CONTENT, PCO2_CONTENT, respectively) are obtained from the CANYON-B and CONTENT methods (Sauzède et al. 2017; Bittig et al. 2018). CANYON-B neural-network-based method was trained using the high quality nutrient (NO_3^-, PO_4^{3-} and $Si(OH)_4$) and carbonate system variables (TALK, TCO2, pH, and pCO2) data collected over the last 30 years available from the GLODAPv2 database (Olsen et al. 2016). CANYON-B derives the nutrients and carbonate system variables from concomitant measurements of pressure, temperature, salinity and O2 associated with the geolocation and date of sampling. CONTENT uses the CANYON-B derived carbonate system variables and derives more robust carbonate system variables. CONTENT combines four carbonate system variables to give a consistent state of the carbonate system.

For the Mediterranean Sea, the nutrients and carbonate system variables are obtained from CANYON-MED method, a dedicated regional version of CANYON-B (Fourrier et al. 2020). This regional neural network-based method was trained using a database specifically gathered for the Mediterranean Sea.

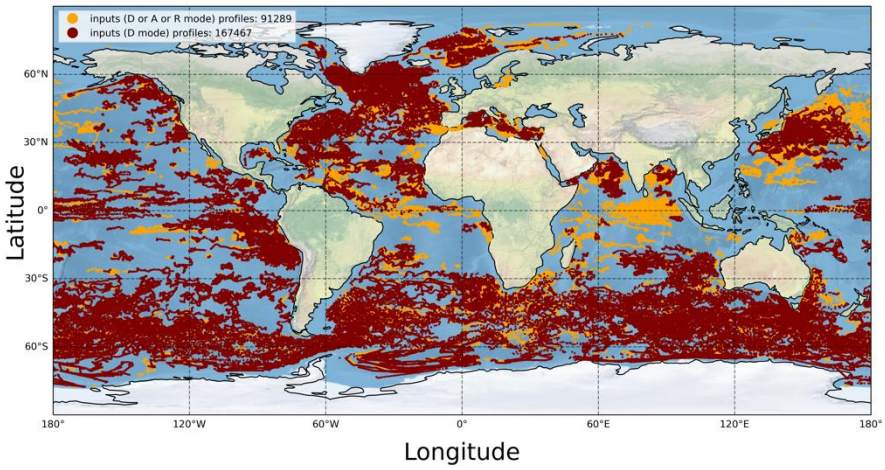
Based on CANYON methods (i.e. CANYON-B, CONTENT and CANYON-MED), the vertical distribution of nutrient and carbonate system variables are estimated from each concomitant BGC-Argo pressure, temperature, salinity and O_2 measurement available in the BGC-Argo synthetic NetCDF files (with a quality control flag equal to 1, 2 or 8, see the Argo reference table 2 for details on quality control flags in https://doi.org/10.13155/29825). The BGC-Argo system

offers three data streams associated with each parameter: the real-time (R-mode), the real-time-adjusted (A-mode), delayed-mode (D-mode). R-mode provides the raw data alongside with the basic quality checks, A-mode provides data automatically adjusted and qualified, and D-mode provides highest accuracy profiles after thorough comprehensive post-processing and quality control including manual inspection by either the float's PI or Delayed Mode QC expert (Bittig et al. 2019).

The pressure, temperature, and salinity profiles used for extracting CANYON/CONTENT estimations originate from real-time, adjusted in real-time, or delayed mode qualified data (i.e. R, A or D-mode), whereas for O_2, only adjusted in real-time or delayed mode (i.e. A or D-mode) qualified data are utilized. A sensitivity analysis was performed to evaluate the impact of using less accurate input data in CANYON/CONTENT. The results indicate that the errors associated with temperature and salinity inputs in R or A-mode are negligible compared to D-mode data. However, for O_2, the errors associated with A-mode data are significantly higher than those in D-mode, as detailed in the QUID. These errors are incorporated into the uncertainties provided in the product. Thus, R-mode temperature and salinity are used as input when A or D-mode profiles are not available, in conjunction with A or D-mode profiles of O_2 for the CANYON/CONTENT derived products.

The use of nutrients and carbonate system variables profiles derived from CANYON/CONTENT using qualified profiles as inputs (i.e., temperature, salinity and O_2 qualified in D-mode) is recommended. Users can use other profiles as well, but it is worth noting that the use of O_2 A-mode as input introduces additional errors in CANYON/CONTENT estimations. The QUID documents the global and regional error propagation into CANYON/CONTENT products.

The error of estimation is available together with each retrieval (NITRATE_CANYON_ERROR, PHOSPHATE_CANYON_ERROR, SILICATE_CANYON_ERROR, TALK_CONTENT_ERROR, TCO2_CONTENT_ERROR, PH_CONTENT_ERROR, PCO2_CONTENT_ERROR for the NO_3^-, PO_4^{3-} , $Si(OH)_4$, TALK, TCO2, pH and pCO_2, respectively). This error is refined depending on the data mode used as inputs based on the sensitivity analysis presented in the QUID.Note that the profiles were not produced for the Black Sea because the global CANYON-B and CONTENT methods are not adapted to this specific area and no regional method dedicated to the Black Sea has been developed so far. The geographical locations of all nutrients and carbon profiles extracted for making this document, accessed from the GDAC on April 4 2024, are shown in Figure 1.

Domain	GLOBAL (180°W-180°E ; 90°S – 90°N)
Resolution	BGC-Argo O_2 profiles resolution
Geographic coverage	This product is global with the spatial and temporal resolution of the BGC-Argo O_2 dataset. A total number of 258,756profiles (November 2024 release; data accessed from GDAC on 04/04/2024).  *Figure 1: Geographical distributions of CANYON/CONTENT profiles (including any of the input in real-time data in orange and all the inputs in delayed mode qualified data in brown), accessed from the GDAC on 04/04/2024. Out of a total of 258,756 profiles, 167,467 are classified as qualified in delayed mode profiles (similar to previous releases).*

II.4 Grid

Discrete, corresponding to BGC-Argo O_2 profiles horizontal resolution

II.5 Vertical Levels

From 70 to 5044 (depending on the vertical levels of the Argo profiles)

II.6 Processing information

II.6.1 Update Time

The product will be updated on a monthly basis with new additional profiles up to one month (M-1) before the current month (M). Please note that the number of profiles in the monthly files may vary whenever there is a change in the data mode of input mainly O_2 (real-time to adjusted in real-time or delayed mode).

II.6.2 Temporal extent stored on delivery mechanism

From September 2002 to M-1 (see product improvements pages https://marine.copernicus.eu/user-corner/user-notification-service) /).

II.6.3 Time averaging

No time averaging, the profiles are aggregated in monthly NetCDF with all available profiles for each month in the associated NetCDF file.

III FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:

What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

IV.1 Nomenclature of original files

MULTIOBS_GLO_BGC_NUTRIENTS_CARBON_PROFILES_MYNRT_015_009 files nomenclature when downloaded through the CMEMS FTP service is based on dataset name and field date.

The scheme is: **cmems_obs-mob_glob_bgc-nut-car_mynrt_irr_i_*${date1}*.nc**

where:

- **date1**: integer (format YYYYMM) corresponding to the field date
- **.nc**: standard NetCDF filename extension.

Example: cmems_obs-mob_glob_bgc-nut-car_mynrt_irr_i_201612.nc (December 2016 field)

IV.2 Index file

An index file is provided containing essential information for changing profiles, including file name, longitude and latitude of the profile, WMO, cycle number, date of the profile, data modes for temperature, salinity, and O_2, as well as the date of update of one of the inputs. This feature will enable users to select the most suitable profiles for their needs. Index file is useful for regular synchronization between DU and operational users.

The index file header comprises several lines beginning with the "#" character, containing metadata about the project, dataset content and date of update. Each line corresponds to a profile with WMO and cycle information. The index lines are sorted based on the date and time of the float measurements. Each field is separated by a comma (","). This enables users to easily access all CANYON/CONTENT variable (for example for NITRATE_CANYON[N_PROF, N_LEVELS]) profile from the monthly file using the WMO (PLATFORM_NUMBER[N_PROF,]) and cycle information (CYCLE_NUMBER[N_PROF,]).

Maintaining consistency between the index file and the profiles within the file system is crucial. It is imperative that profiles listed in the index file also exist within the monthly files. The index file will be updated on a monthly basis with new additions of profiles. The number of profiles in the monthly files can change, whenever there is a change in the data mode of input mainly O_2 (real-time to adjusted in real-time or delayed mode). These changes will be promptly reflected in the index file.

Copernicus CANYON data index file example
title : Nutrient and Carbon profiles vertical distribution
project : CMEMS - MULTIOBS Thematic Assembly Center
product_id : MULTIOBS_GLO_BGC_NUTRIENTS_CARBON_PROFILES_MYNRT_015_009
dataset_id : cmems_obs-mob_glo_bgc-nut-car_mynrt_irr_i
date of update: 20240514144110
file_name,date,wmo,n_cycle,longitude,latitude,temp_dt_mode,psal_dt_mode,doxy_dt_mode,date_update
2002/cmems_obs-mob_glo_bgc-nut-car_mynrt_irr_i_200209.nc,2002-09-08 23:02:51,4900093,1,-157.987,22.882,D,D,D,20240507
2002/cmems_obs-mob_glo_bgc-nut-car_mynrt_irr_i_200209.nc,2002-09-19 12:00:41,4900093,2,-157.812,22.91,D,D,D,20240507
2002/cmems_obs-mob_glo_bgc-nut-car_mynrt_irr_i_200209.nc,2002-09-30 01:30:00,4900093,3,-157.416,22.887,D,D,D,20240507
2002/cmems_obs-mob_glo_bgc-nut-car_mynrt_irr_i_200210.nc,2002-10-10 14:18:32,4900093,4,-157.025,22.803,D,D,D,20240507
2002/cmems_obs-mob_glo_bgc-nut-car_mynrt_irr_i_200210.nc,2002-10-21 03:05:10,4900093,5,-156.831,22.736,D,D,D,20240507

IV.3 Other information: mean centre of Products, land mask value, missing value

The missing values are equal to "_FillValue" (see variable attribute on NetCDF files).

IV.3.1 File size

Table 3: dataset size information table.

PRODUCT NAME	DIMENSION [GB] Full dataset	
	Compressed	**Uncompressed**
MULTIOBS_GLO_**BGC**_NUTRIENTS_CARBON_PROFILES_MYNRT_015_009	5.0	60.0

IV.4 Structure of files

Examples of the header of output NetCDF files are inserted in annex, for each dataset.

V REFERENCES

Bittig, Henry C., Tanya L. Maurer, Josh N. Plant, Annie P. Wong, Catherine Schmechtig, Hervé Claustre, Thomas W. Trull, et al. 2019. "A BGC-Argo Guide: Planning, Deployment, Data Handling and Usage." *Frontiers in Marine Science*. https://doi.org/10.3389/fmars.2019.00502.

Bittig, Henry C., Tobias Steinhoff, Hervé Claustre, Björn Fiedler, Nancy L. Williams, Raphaëlle Sauzède, Arne Körtzinger, and Jean Pierre Gattuso. 2018. "An Alternative to Static Climatologies: Robust Estimation of Open Ocean CO2 Variables and Nutrient Concentrations from T, S, and O2 Data Using Bayesian Neural Networks." *Frontiers in Marine Science* 5 (SEP): 1–29. https://doi.org/10.3389/fmars.2018.00328.

Fourrier, Marine, Laurent Coppola, Hervé Claustre, Fabrizio D'Ortenzio, Raphaëlle Sauzède, and Jean Pierre Gattuso. 2020. "A Regional Neural Network Approach to Estimate Water-Column Nutrient Concentrations and Carbonate System Variables in the Mediterranean Sea: CANYON-MED." *Frontiers in Marine Science* 7 (August): 1–20. https://doi.org/10.3389/fmars.2020.00620.

Olsen, Are, Robert M. Key, Steven Van Heuven, Siv K. Lauvset, Anton Velo, Xiaohua Lin, Carsten Schirnick, et al. 2016. "The Global Ocean Data Analysis Project Version 2 (GLODAPv2) - An Internally Consistent Data Product for the World Ocean." *Earth System Science Data* 8 (2): 297–323. https://doi.org/10.5194/essd-8-297-2016.

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DR 1 Quality Information Document (QUID) :
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-MOB-QUID-015-009.pdf

VI ANNEX

```
netcdf cmems_obs-mob_glo_bgc-nut-car_mynrt_irr_i_201612 {
dimensions:
        N_LEVELS = 2293 ;
        N_PROF = 1282 ;
        DATE_TIME = 14 ;
        STRING2 = 2 ;
        STRING8 = 8 ;
        STRING64 = 64 ;
        N_PARAM = 25 ;
        STRING16 = 16 ;
variables:
        float PRES_ADJUSTED(N_PROF, N_LEVELS) ;
                PRES_ADJUSTED:units = "decibar" ;
                PRES_ADJUSTED:_FillValue = 99999.f ;
                PRES_ADJUSTED:long_name = "Sea Pressure" ;
                PRES_ADJUSTED:standard_name = "sea_water_pressure" ;
                PRES_ADJUSTED:comment = "In situ measurement" ;
                PRES_ADJUSTED:C_format = "%7.1f" ;
                PRES_ADJUSTED:FORTRAN_format = "F7.1" ;
                PRES_ADJUSTED:valid_min = 0. ;
                PRES_ADJUSTED:valid_max = 12000. ;
                PRES_ADJUSTED:resolution = "0.1f" ;
        char PRES_ADJUSTED_QC(N_PROF, N_LEVELS) ;
                PRES_ADJUSTED_QC:_FillValue = " " ;
                PRES_ADJUSTED_QC:long_name = "quality flag" ;
                PRES_ADJUSTED_QC:conventions = "Argo reference table 2" ;
        float PRES_ADJUSTED_ERROR(N_PROF, N_LEVELS) ;
                PRES_ADJUSTED_ERROR:units = "decibar" ;
                PRES_ADJUSTED_ERROR:_FillValue = 99999.f ;
```

PRES_ADJUSTED_ERROR:long_name = "Sea water pressure, equals 0 at sea-level" ;

PRES_ADJUSTED_ERROR:comment = "Contains the error on the adjusted values as determined by the delayed mode QC process." ;

PRES_ADJUSTED_ERROR:C_format = "%7.1f" ;

PRES_ADJUSTED_ERROR:FORTRAN_format = "F7.1" ;

PRES_ADJUSTED_ERROR:resolution = "0.1f" ;

float TEMP_ADJUSTED(N_PROF, N_LEVELS) ;

TEMP_ADJUSTED:units = "degree_Celsius" ;

TEMP_ADJUSTED:_FillValue = 99999.f ;

TEMP_ADJUSTED:long_name = "Sea temperature in-situ ITS-90 scale" ;

TEMP_ADJUSTED:standard_name = "sea_water_temperature" ;

TEMP_ADJUSTED:comment = "In situ masurement" ;

TEMP_ADJUSTED:C_format = "%9.3f" ;

TEMP_ADJUSTED:FORTRAN_format = "F9.3" ;

TEMP_ADJUSTED:valid_min = -2.5 ;

TEMP_ADJUSTED:valid_max = 40. ;

TEMP_ADJUSTED:resolution = "0.001f" ;

char TEMP_ADJUSTED_QC(N_PROF, N_LEVELS) ;

TEMP_ADJUSTED_QC:_FillValue = " " ;

TEMP_ADJUSTED_QC:long_name = "quality flag" ;

TEMP_ADJUSTED_QC:conventions = "Argo reference table 2" ;

float TEMP_ADJUSTED_ERROR(N_PROF, N_LEVELS) ;

TEMP_ADJUSTED_ERROR:units = "degree_Celsius" ;

TEMP_ADJUSTED_ERROR:_FillValue = 99999.f ;

TEMP_ADJUSTED_ERROR:long_name = "Sea temperature in-situ ITS-90 scale" ;

TEMP_ADJUSTED_ERROR:comment = "Contains the error on the adjusted values as determined by the delayed mode QC process." ;

TEMP_ADJUSTED_ERROR:C_format = "%9.3f" ;

TEMP_ADJUSTED_ERROR:FORTRAN_format = "F9.3" ;

TEMP_ADJUSTED_ERROR:resolution = "0.001f" ;

float PSAL_ADJUSTED(N_PROF, N_LEVELS) ;

PSAL_ADJUSTED:units = "0.001" ;

PSAL_ADJUSTED:_FillValue = 99999.f ;

PSAL_ADJUSTED:long_name = "Practical Salinity" ;

PSAL_ADJUSTED:standard_name = "sea_water_salinity" ;

PSAL_ADJUSTED:comment = "In situ masurement" ;

PSAL_ADJUSTED:C_format = "%9.3f" ;

PSAL_ADJUSTED:FORTRAN_format = "F9.3" ;

PSAL_ADJUSTED:valid_min = 2. ;

PSAL_ADJUSTED:valid_max = 41. ;

PSAL_ADJUSTED:resolution = "0.001f" ;

PSAL_ADJUSTED:unit_long = "practical salinity unit" ;

char PSAL_ADJUSTED_QC(N_PROF, N_LEVELS) ;

PSAL_ADJUSTED_QC:_FillValue = " " ;

PSAL_ADJUSTED_QC:long_name = "quality flag" ;

PSAL_ADJUSTED_QC:conventions = "Argo reference table 2" ;

float PSAL_ADJUSTED_ERROR(N_PROF, N_LEVELS) ;

PSAL_ADJUSTED_ERROR:units = "0.001" ;

PSAL_ADJUSTED_ERROR:_FillValue = 99999.f ;

PSAL_ADJUSTED_ERROR:long_name = "Practical Salinity" ;

PSAL_ADJUSTED_ERROR:comment = "Contains the error on the adjusted values as determined by the delayed mode QC process." ;

PSAL_ADJUSTED_ERROR:C_format = "%9.3f" ;

PSAL_ADJUSTED_ERROR:FORTRAN_format = "F9.3" ;

PSAL_ADJUSTED_ERROR:resolution = "0.001f" ;

PSAL_ADJUSTED_ERROR:unit_long = "practical salinity unit" ;

float DOXY_ADJUSTED(N_PROF, N_LEVELS) ;

DOXY_ADJUSTED:units = "micromole kg-1" ;

DOXY_ADJUSTED:_FillValue = 99999.f ;

DOXY_ADJUSTED:long_name = "Dissolved oxygen" ;

DOXY_ADJUSTED:standard_name = "moles_of_oxygen_per_unit_mass_in_sea_water" ;

DOXY_ADJUSTED:comment = "In situ masurement" ;

DOXY_ADJUSTED:C_format = "%9.3f" ;

DOXY_ADJUSTED:FORTRAN_format = "F9.3" ;

DOXY_ADJUSTED:valid_min = -5. ;

DOXY_ADJUSTED:valid_max = 600. ;

DOXY_ADJUSTED:resolution = "0.001f" ;

char DOXY_ADJUSTED_QC(N_PROF, N_LEVELS) ;

DOXY_ADJUSTED_QC:_FillValue = " " ;

DOXY_ADJUSTED_QC:long_name = "quality flag" ;

DOXY_ADJUSTED_QC:conventions = "Argo reference table 2" ;

float DOXY_ADJUSTED_ERROR(N_PROF, N_LEVELS) ;

DOXY_ADJUSTED_ERROR:units = "micromole kg-1" ;

DOXY_ADJUSTED_ERROR:_FillValue = 99999.f ;

DOXY_ADJUSTED_ERROR:long_name = "Dissolved oxygen" ;

DOXY_ADJUSTED_ERROR:comment = "Contains the error on the adjusted values as determined by the delayed mode QC process." ;

DOXY_ADJUSTED_ERROR:C_format = "%9.3f" ;

DOXY_ADJUSTED_ERROR:FORTRAN_format = "F9.3" ;

DOXY_ADJUSTED_ERROR:resolution = "0.001f" ;

float NITRATE_CANYON(N_PROF, N_LEVELS) ;

NITRATE_CANYON:units = "micromole kg-1" ;

NITRATE_CANYON:_FillValue = 99999.f ;

NITRATE_CANYON:long_name = "Nitrate" ;

NITRATE_CANYON:standard_name = "moles_of_nitrate_per_unit_mass_in_sea_water" ;

NITRATE_CANYON:comment = "CANYON-derived estimations" ;

NITRATE_CANYON:C_format = "%9.3f" ;

NITRATE_CANYON:FORTRAN_format = "F9.3" ;

NITRATE_CANYON:resolution = "0.001f" ;

char NITRATE_CANYON_QC(N_PROF, N_LEVELS) ;

NITRATE_CANYON_QC:_FillValue = " " ;

NITRATE_CANYON_QC:long_name = "quality flag" ;

NITRATE_CANYON_QC:conventions = "Argo reference table 2" ;

float NITRATE_CANYON_ERROR(N_PROF, N_LEVELS) ;

NITRATE_CANYON_ERROR:units = "micromole kg-1" ;

NITRATE_CANYON_ERROR:_FillValue = 99999.f ;

NITRATE_CANYON_ERROR:long_name = "Nitrate" ;

NITRATE_CANYON_ERROR:comment = "Contains the error on the adjusted values used as inputs in the CANYON method as determined by the delayed mode QC process." ;

NITRATE_CANYON_ERROR:C_format = "%9.3f" ;

NITRATE_CANYON_ERROR:FORTRAN_format = "F9.3" ;

NITRATE_CANYON_ERROR:resolution = "0.001f" ;

float PHOSPHATE_CANYON(N_PROF, N_LEVELS) ;

PHOSPHATE_CANYON:units = "micromole kg-1" ;

PHOSPHATE_CANYON:_FillValue = 99999.f ;

PHOSPHATE_CANYON:long_name = "Phosphate" ;

PHOSPHATE_CANYON:standard_name = "moles_of_phosphate_per_unit_mass_in_sea_water" ;

PHOSPHATE_CANYON:comment = "CANYON-derived estimations " ;

PHOSPHATE_CANYON:C_format = "%9.3f" ;

PHOSPHATE_CANYON:FORTRAN_format = "F9.3" ;

PHOSPHATE_CANYON:resolution = "0.001f" ;

char PHOSPHATE_CANYON_QC(N_PROF, N_LEVELS) ;

PHOSPHATE_CANYON_QC:_FillValue = " " ;

PHOSPHATE_CANYON_QC:long_name = "quality flag" ;

PHOSPHATE_CANYON_QC:conventions = "Argo reference table 2" ;

float PHOSPHATE_CANYON_ERROR(N_PROF, N_LEVELS) ;

PHOSPHATE_CANYON_ERROR:units = "micromole kg-1" ;

PHOSPHATE_CANYON_ERROR:_FillValue = 99999.f ;

PHOSPHATE_CANYON_ERROR:long_name = "Phosphate" ;

PHOSPHATE_CANYON_ERROR:comment = "Contains the error on the adjusted values used as inputs in the CANYON method as determined by the delayed mode QC process." ;

PHOSPHATE_CANYON_ERROR:C_format = "%9.3f" ;

PHOSPHATE_CANYON_ERROR:FORTRAN_format = "F9.3" ;

PHOSPHATE_CANYON_ERROR:resolution = "0.001f" ;

float SILICATE_CANYON(N_PROF, N_LEVELS) ;

SILICATE_CANYON:units = "micromole kg-1" ;

SILICATE_CANYON:_FillValue = 99999.f ;

SILICATE_CANYON:long_name = "Silicate" ;

SILICATE_CANYON:standard_name = "moles_of_silicate_per_unit_mass_in_sea_water" ;

SILICATE_CANYON:comment = "CANYON-derived estimations" ;

SILICATE_CANYON:C_format = "%9.3f" ;

SILICATE_CANYON:FORTRAN_format = "F9.3" ;

SILICATE_CANYON:resolution = "0.001f" ;

char SILICATE_CANYON_QC(N_PROF, N_LEVELS) ;

SILICATE_CANYON_QC:_FillValue = " " ;

SILICATE_CANYON_QC:long_name = "quality flag" ;

SILICATE_CANYON_QC:conventions = "Argo reference table 2" ;

float SILICATE_CANYON_ERROR(N_PROF, N_LEVELS) ;

SILICATE_CANYON_ERROR:units = "micromole kg-1" ;

SILICATE_CANYON_ERROR:_FillValue = 99999.f ;

SILICATE_CANYON_ERROR:long_name = "Silicate" ;

SILICATE_CANYON_ERROR:comment = "Contains the error on the adjusted values used as inputs in the CANYON method as determined by the delayed mode QC process." ;

SILICATE_CANYON_ERROR:C_format = "%9.3f" ;

SILICATE_CANYON_ERROR:FORTRAN_format = "F9.3" ;

SILICATE_CANYON_ERROR:resolution = "0.001f" ;

float TALK_CONTENT(N_PROF, N_LEVELS) ;

TALK_CONTENT:units = "micromole kg-1" ;

TALK_CONTENT:_FillValue = 99999.f ;

TALK_CONTENT:long_name = "Total alkalinity" ;

TALK_CONTENT:standard_name = "total_alkalinity_in_sea_water" ;

TALK_CONTENT:comment = "CONTENT-derived estimations" ;

TALK_CONTENT:C_format = "%9.3f" ;

TALK_CONTENT:FORTRAN_format = "%9.3f" ;

TALK_CONTENT:resolution = "0.001f" ;

char TALK_CONTENT_QC(N_PROF, N_LEVELS) ;

TALK_CONTENT_QC:_FillValue = " " ;

TALK_CONTENT_QC:long_name = "quality flag" ;

TALK_CONTENT_QC:conventions = "Argo reference table 2" ;

float TALK_CONTENT_ERROR(N_PROF, N_LEVELS) ;

TALK_CONTENT_ERROR:units = "micromole kg-1" ;

TALK_CONTENT_ERROR:_FillValue = 99999.f ;

TALK_CONTENT_ERROR:long_name = "Total alkalinity" ;

TALK_CONTENT_ERROR:comment = "Contains the error on the adjusted values used as inputs in the CONTENT method as determined by the delayed mode QC process." ;

TALK_CONTENT_ERROR:C_format = "%9.3f" ;

TALK_CONTENT_ERROR:FORTRAN_format = "%9.3f" ;

TALK_CONTENT_ERROR:resolution = "0.001f" ;

float TCO2_CONTENT(N_PROF, N_LEVELS) ;

TCO2_CONTENT:units = "micromole kg-1" ;

TCO2_CONTENT:_FillValue = 99999.f ;

TCO2_CONTENT:long_name = "Dissolved Inorganic Carbon" ;

TCO2_CONTENT:standard_name = "dissolved_inorganic_carbon_in_sea_water" ;

TCO2_CONTENT:comment = "CONTENT-derived estimations" ;

TCO2_CONTENT:C_format = "%9.3f" ;

TCO2_CONTENT:FORTRAN_format = "%9.3f" ;

TCO2_CONTENT:resolution = "0.001f" ;

char TCO2_CONTENT_QC(N_PROF, N_LEVELS) ;

TCO2_CONTENT_QC:_FillValue = " " ;

TCO2_CONTENT_QC:long_name = "quality flag" ;

TCO2_CONTENT_QC:conventions = "Argo reference table 2" ;

float TCO2_CONTENT_ERROR(N_PROF, N_LEVELS) ;

TCO2_CONTENT_ERROR:units = "micromole kg-1" ;

TCO2_CONTENT_ERROR:_FillValue = 99999.f ;

TCO2_CONTENT_ERROR:long_name = "Dissolved Inorganic Carbon" ;

TCO2_CONTENT_ERROR:comment = "Contains the error on the adjusted values used as inputs in the CONTENT method as determined by the delayed mode QC process." ;

TCO2_CONTENT_ERROR:C_format = "%9.3f" ;

TCO2_CONTENT_ERROR:FORTRAN_format = "%9.3f" ;

TCO2_CONTENT_ERROR:resolution = "0.001f" ;

float PH_CONTENT(N_PROF, N_LEVELS) ;

PH_CONTENT:units = "no unit" ;

PH_CONTENT:_FillValue = 99999.f ;

PH_CONTENT:long_name = "pH" ;

PH_CONTENT:standard_name = "sea_water_ph_reported_on_total_scale" ;

PH_CONTENT:comment = "CONTENT-derived estimations" ;

PH_CONTENT:C_format = "%9.3f" ;

PH_CONTENT:FORTRAN_format = "%9.3f" ;

PH_CONTENT:resolution = "0.001f" ;

char PH_CONTENT_QC(N_PROF, N_LEVELS) ;

PH_CONTENT_QC:_FillValue = " " ;

PH_CONTENT_QC:long_name = "quality flag" ;

PH_CONTENT_QC:conventions = "Argo reference table 2" ;

float PH_CONTENT_ERROR(N_PROF, N_LEVELS) ;

PH_CONTENT_ERROR:units = "no unit" ;

PH_CONTENT_ERROR:_FillValue = 99999.f ;

PH_CONTENT_ERROR:long_name = "pH" ;

PH_CONTENT_ERROR:comment = "Contains the error on the adjusted values used as inputs in the CONTENT method as determined by the delayed mode QC process." ;

PH_CONTENT_ERROR:C_format = "%9.3f" ;

PH_CONTENT_ERROR:FORTRAN_format = "%9.3f" ;

PH_CONTENT_ERROR:resolution = "0.001f" ;

float PCO2_CONTENT(N_PROF, N_LEVELS) ;

PCO2_CONTENT:units = "micro atm" ;

PCO2_CONTENT:_FillValue = 99999.f ;

PCO2_CONTENT:long_name = "Partial pressure of carbon dioxide" ;

PCO2_CONTENT:standard_name = "partial_pressure_of_carbon_dioxide_in_sea_water" ;

PCO2_CONTENT:comment = "CONTENT-derived estimations" ;

PCO2_CONTENT:C_format = "%9.3f" ;

PCO2_CONTENT:FORTRAN_format = "%9.3f" ;

PCO2_CONTENT:resolution = "0.001f" ;

```
char PCO2_CONTENT_QC(N_PROF, N_LEVELS) ;
        PCO2_CONTENT_QC:_FillValue = " " ;
        PCO2_CONTENT_QC:long_name = "quality flag" ;
        PCO2_CONTENT_QC:conventions = "Argo reference table 2" ;
float PCO2_CONTENT_ERROR(N_PROF, N_LEVELS) ;
        PCO2_CONTENT_ERROR:units = "micro atm" ;
        PCO2_CONTENT_ERROR:_FillValue = 99999.f ;
        PCO2_CONTENT_ERROR:long_name = "Partial pressure of carbon dioxide" ;
        PCO2_CONTENT_ERROR:comment = "Contains the error on the adjusted values
used as inputs in the CONTENT method as determined by the delayed mode QC process." ;
        PCO2_CONTENT_ERROR:C_format = "%9.3f" ;
        PCO2_CONTENT_ERROR:FORTRAN_format = "%9.3f" ;
        PCO2_CONTENT_ERROR:resolution = "0.001f" ;
char REFERENCE_DATE_TIME(DATE_TIME) ;
        REFERENCE_DATE_TIME:_FillValue = " " ;
        REFERENCE_DATE_TIME:long_name = "Date of reference for Julian days" ;
        REFERENCE_DATE_TIME:comment = "Date of reference for Julian days" ;
        REFERENCE_DATE_TIME:conventions = "YYYYMMDDHHMISS" ;
double LONGITUDE(N_PROF) ;
        LONGITUDE:units = "degree_east" ;
        LONGITUDE:_FillValue = 99999. ;
        LONGITUDE:long_name = "Longitude of the station, best estimate" ;
        LONGITUDE:valid_min = -180. ;
        LONGITUDE:valid_max = 180. ;
double LATITUDE(N_PROF) ;
        LATITUDE:units = "degree_north" ;
        LATITUDE:_FillValue = 99999. ;
        LATITUDE:long_name = "Latitude of the station, best estimate" ;
        LATITUDE:valid_min = -90. ;
        LATITUDE:valid_max = 90. ;
char POSITION_QC(N_PROF, STRING2) ;
        POSITION_QC:_FillValue = " " ;
```

POSITION_QC:long_name = "Quality on position (latitude and longitude)" ;

POSITION_QC:conventions = "Argo reference table 2" ;

double JULD(N_PROF) ;

JULD:units = "days since 1950-01-01 00:00:00 UTC" ;

JULD:_FillValue = 999999. ;

JULD:long_name = "Julian day (UTC) of the station relative to REFERENCE_DATE_TIME" ;

JULD:conventions = "Relative julian days with decimal part (as parts of day)" ;

char JULD_QC(N_PROF, STRING2) ;

JULD_QC:_FillValue = " " ;

JULD_QC:long_name = "Quality on Date and Time" ;

JULD_QC:conventions = "Argo reference table 2" ;

char PLATFORM_NUMBER(N_PROF, STRING8) ;

PLATFORM_NUMBER:_FillValue = " " ;

PLATFORM_NUMBER:long_name = "Float unique identifier" ;

PLATFORM_NUMBER:conventions = "WMO float identifier : A9IIIII" ;

int CYCLE_NUMBER(N_PROF) ;

CYCLE_NUMBER:_FillValue = 99999 ;

CYCLE_NUMBER:long_name = "Float cycle number" ;

CYCLE_NUMBER:conventions = "0..N, 0 : launch cycle (if exists), 1 : first complete cycle" ;

char PI_NAME(N_PROF, STRING64) ;

PI_NAME:long_name = "Name of the principal investigator" ;

PI_NAME:comment = "Name of the principal investigator" ;

char STATION_PARAMETERS(N_PROF, N_PARAM, STRING64) ;

STATION_PARAMETERS:_FillValue = " " ;

STATION_PARAMETERS:long_name = "List of available parameters for the station" ;

STATION_PARAMETERS:conventions = "Argo reference table 3" ;

char WMO_INST_TYPE(N_PROF, STRING16) ;

WMO_INST_TYPE:_FillValue = " " ;

WMO_INST_TYPE:long_name = "Coded instrument type" ;

WMO_INST_TYPE:conventions = "Argo reference table 8" ;

char DATA_CENTRE(N_PROF, STRING2) ;

DATA_CENTRE:_FillValue = " " ;

DATA_CENTRE:long_name = "Data centre in charge of float data processing" ;

DATA_CENTRE:conventions = "Argo reference table 4" ;

// global attributes:

:title = "Nutrient and Carbon profiles vertical distribution" ;

:project = "CMEMS - MULTIOBS Thematic Assembly Center" ;

:project_url = "http://marine.copernicus.eu" ;

:contact = "servicedesk.cmems@mercator-ocean.eu" ;

:description = "CMEMS MULTIOBS vertical profiles of the concentration of nitrates, phosphates and silicates, Total alkalinity, Dissolved inorganic carbon, pH, and PCO2 computed for each Argo float equipped with an oxygen sensor using the CANYON-B and CONTENT Neural Network-based method at the global scale and using the CANYON-MED Neural Network-based method for the Mediterranean Sea." ;

:Conventions = "CF-1.6" ;

:institution = "IMEV/LOV/CNRS" ;

:domain_name = "GLO" ;

:references = "Bittig et al. 2018 (Frontiers in Marine Science; https://doi.org/10.3389/fmars.2018.00328); Fourrier et al. 2020 (Frontiers in Marine Science; https://doi.org/10.3389/fmars.2020.00620); Sauzede et al. 2017 (Frontiers in Marine Science; https://doi.org/10.3389/fmars.2017.00128)" ;

}